American Computer Science League

2024-2025 ● Contest 3: Manhattan Distance ● Intermediate Division

PROBLEM: In a *Zig-Zag Triangle* pattern, numbers are written in rows that increase in length, and the numbers in the rows alternate being written from left-to-right and right-to-left. The amount by which the rows increase in length is called the pattern's *delta*. In the following example, the delta value is 2:

```
 1
 4  3  2
 5  6  7  8  9
16 15 14 13 12 11 10
17 18 19 20 21 22 23 24 25
36 35 34 33 32 31 30 29 28 27 26
....
```

The *Manhattan Distance* between two numbers, *M* and *N*, in a Zig-Zag Triangle is how far the numbers are from each other in both the horizontal and vertical directions. You'll be given a delta value and two numbers, *M* and *N,* that are in the triangle. Your program must find the Manhattan Distance between the numbers *M* and *N*.

EXAMPLE:

Input	`2 17 3`
Output	`4`
Explanation	The delta value for the Zig-Zag Triangle is 2; it is shown above. The Manhattan Distance between the numbers 17 and 3 is 4: go horizontally 1 column and vertically 3 rows to go from one number to the other number.

INPUT: Three positive integers: *delta*, *M*, and *N*, each less than or equal to 1,000.

OUTPUT: Find the Manhattan Distance between *M* and *N* in the Zig-Zag Triangle with the specified delta value.

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SAMPLE INPUT *(for readability, all inputs are shown on a single line):*

```
1. 2 17 3
2. 5 24 25
3. 7 40 30
4. 6 5 100
5. 3 200 180
6. 4 100 120
```

SAMPLE OUTPUT:

```
1. 4
2. 1
3. 10
4. 5
5. 20
6. 20
```

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TEST DATA

TEST INPUT *(for readability, the inputs are shown on a single line):*

```
1. 5 44 33
2. 3 20 70
3. 6 100 200
4. 10 20 28
5. 2 200 1000
6. 25 1000 950
```

TEST OUTPUT:

```
1. 9
2. 19
3. 22
4. 8
5. 38
6. 50
```

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PROBLEM STATEMENT:

In a *Zig-Zag Triangle* pattern, numbers are written in rows that increase in length, and the numbers in the rows alternate being written from left-to-right and right-to-left. The amount by which the rows increase in length is called the pattern's *delta*. In the following example, the delta value is 2:

```
  1
  4  3  2
  5  6  7  8  9
 16 15 14 13 12 11 10
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 ....
```

The *Manhattan Distance* between two numbers, *M* and *N*, in a Zig-Zag Triangle is how far the numbers are from each other in both the horizontal and vertical directions. You'll be given a delta value and two numbers, *M* and *N,* that are in the triangle. Your program must find the Manhattan Distance between the numbers *M* and *N*.

EXAMPLE:

Input:

```
2 17 3
```

Output:

```
4
```

Explanation:

The delta value for the Zig-Zag Triangle is 2; it is shown above. The Manhattan Distance between the numbers 17 and 3 is 4: go horizontally 1 column and vertically 3 rows to go from one number to the other number.

TASK:

Complete the function **findManDist**

- The function has 3 parameters, all positive integers: *delta*, the increase in the width of each row in the triangle and the two numbers, *M* and *N*, representing the 2 numbers to use in finding the Manhattan Distance.
- The function returns an integer for the Manhattan Distance between *M* and *N* in the resulting triangle.

You may create additional functions that are called from **findManDist** if needed in solving the problem.

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CONSTRAINTS:
Each input value is no more than 1,000.

DATA PROVIDED:
There are 6 sets of Sample Data for debugging and 6 sets of Test Data for scoring. The test cases will vary in difficulty. You should create sample data of your own to fully test your program.

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问题：在 *Z* ***字****三角形*模式中，数字按行书写，每行的长度逐渐增加，且数字的书写方向在每行之间交替变化，先从左到右，再从右到左。每行长度增加的幅度称为该模式的*增量*。在以下示例中，增量值为 2：

```
 1
 4  3  2
 5  6  7  8  9
16 15 14 13 12 11 10
17 18 19 20 21 22 23 24 25
36 35 34 33 32 31 30 29 28 27 26
....
```

在 Z **字**三角形中，两个数字 *M* 和 *N* 之间的*曼哈顿距离*是指这两个数字在水平和垂直方向上的距离之和。题目会给出一个增量值以及三角形中的两个数字 *M* 和 *N*。你的程序**需要**计算数字 *M* 和 *N* 之间的曼哈顿距离。

样例：

输入	2 17 3
输出	4
解释说明	该 Z **字**三角形的增量值为 2，如上述示例所示。数字 17 和 3 之间的曼哈顿距离为 4：**从** 17 **到** 3 **需要**水平移动 1 列，垂直移动 3 行。

输入：三个正整数：*增量*、*M* 和 *N*，每个数均小于或等于 1000。

输出：在指定增量值的 Z **字**三角形中，**计算** *M* 和 *N* 之间的曼哈顿距离。

样例输入（*为便于阅读，所有输入均显示在同一行*）：

```
1. 2 17 3
2. 5 24 25
3. 7 40 30
4. 6 5 100
5. 3 200 180
6. 4 100 120
```

样例输出：

```
1. 4
2. 1
3. 10
4. 5
5. 20
6. 20
```

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测试数据

测试输入（*为便于阅读，所有输入均显示在同一行*）：

```
1. 5 44 33
2. 3 20 70
3. 6 100 200
4. 10 20 28
5. 2 200 1000
6. 25 1000 950
```

测试输出：

```
1. 9
2. 19
3. 22
4. 8
5. 38
6. 50
```

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问题陈述：

在 *Z* ***字***三角形模式中，数字按行书写，每行的长度逐渐增加，且数字的书写方向在每行之间交替变化，先从左到右，再从右到左。每行长度增加的幅度称为该模式的*增量*。在以下示例中，增量值为 2：

```
 1
 4  3  2
 5  6  7  8  9
16 15 14 13 12 11 10
17 18 19 20 21 22 23 24 25
36 35 34 33 32 31 30 29 28 27 26
....
```

在 Z **字**三角形中，两个数字 M 和 N 之间的*曼哈顿距离*是指这两个数字在水平和垂直方向上的距离之和。题目会给出一个增量值以及三角形中的两个数字 M 和 N。你的程序**需要**计算数字 M 和 N 之间的曼哈顿距离。

样例：

*输**入**：*

```
2 17 3
```

输出：

```
4
```

*解释**说明**：*

该 Z **字**三角形的增量值为 2，如上述示例所示。数字 17 和 3 之间的曼哈顿距离为 4：**从** 17 **到** 3 **需要**水平移动 1 列，垂直移动 3 行。

任务：

完成**函数** findManDist

- 该函数有 3 个参数，均为正整数：***增量***表示三角形中每行宽度的增加值，M 和 N 表示用于计算曼哈顿距离的两个数字。

- 函数返回一个整数，表示在生成**的**三角形中，M 和 N 之间的曼哈顿距离。

在解决问题的过程中，如有需要，你可以构建额外的函数供 findManDist 调用来解答此问题。

约束条件：

每个输入值不超过 1000。

提供的数据：

有 6 组用于调试的样例数据和 6 组用于评分的测试数据。测试用例的难度各不相同。
你应该创建自己的样例数据来充分测试你的程序。